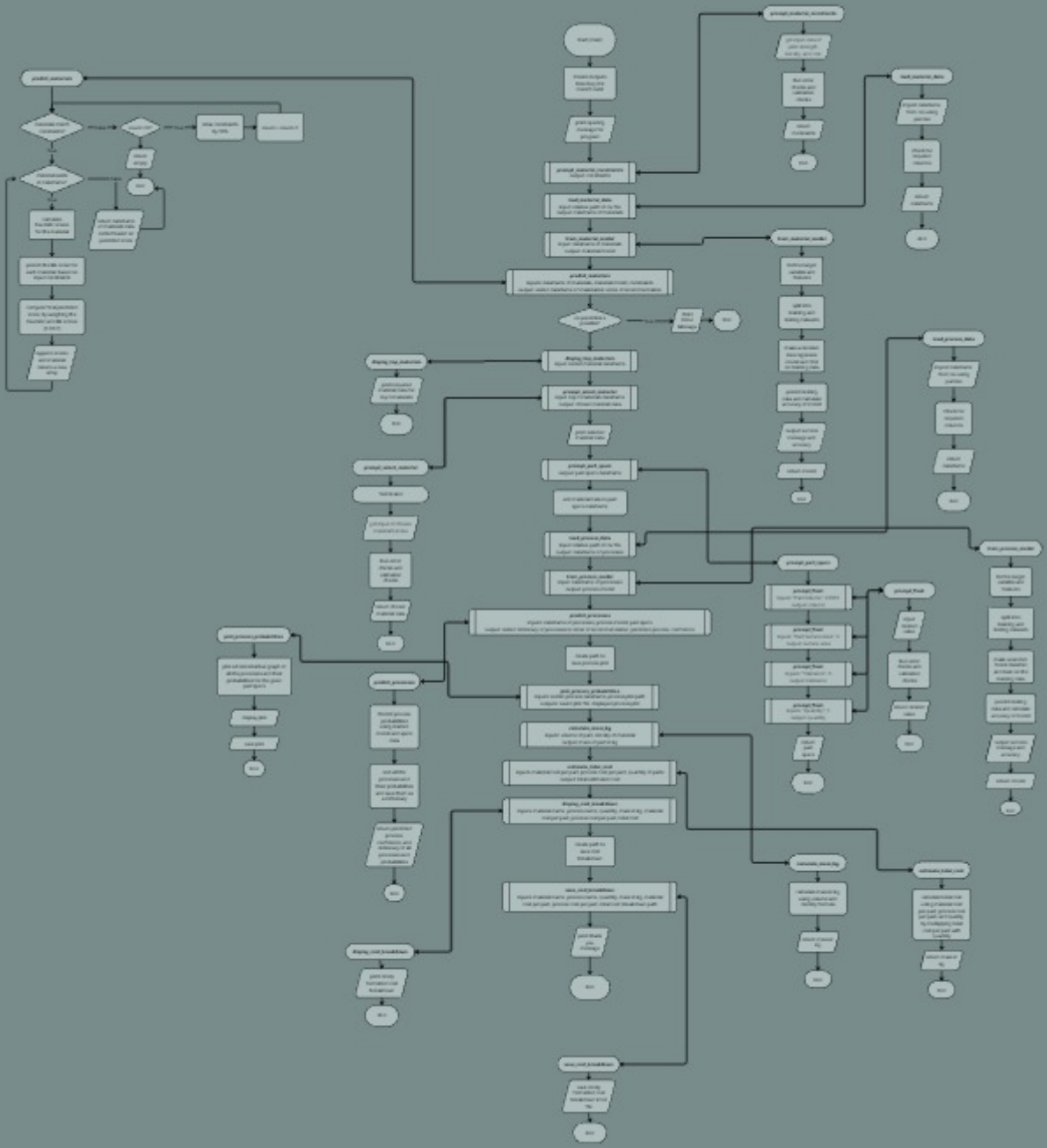
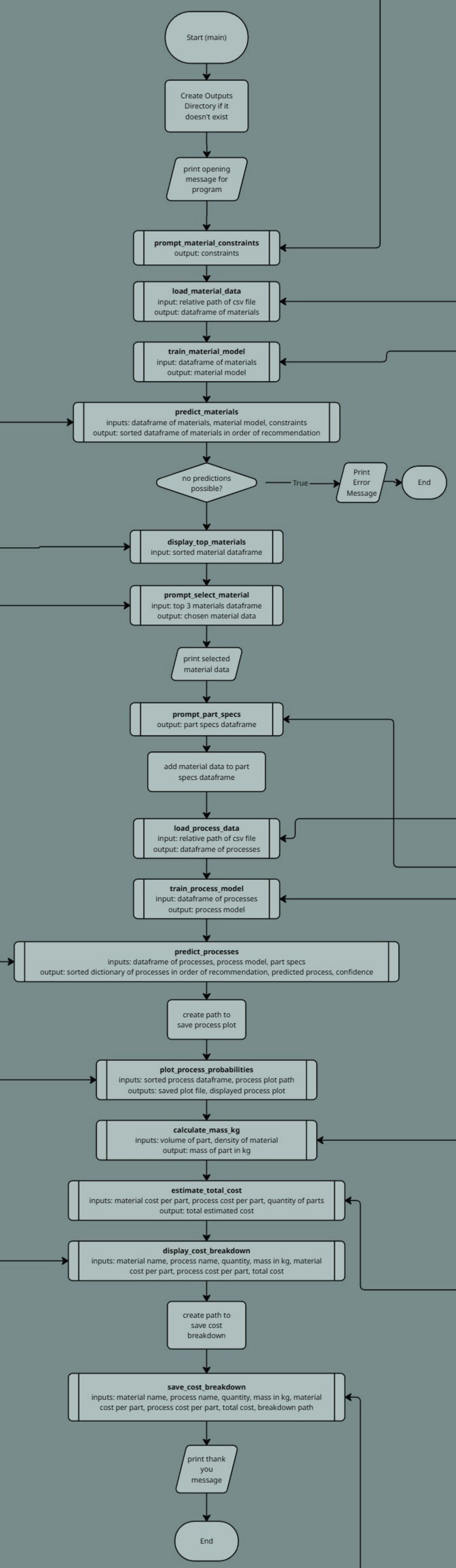
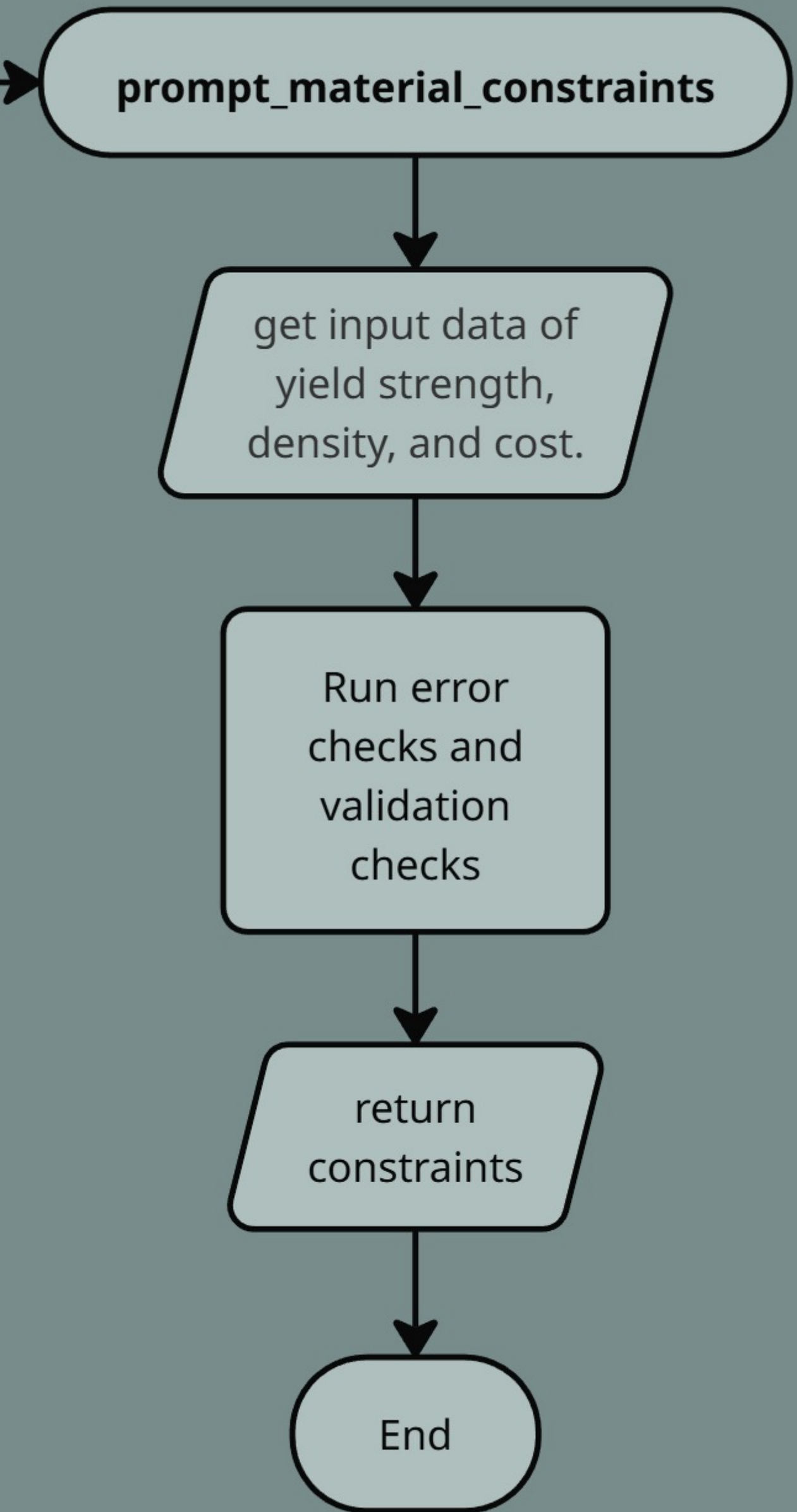
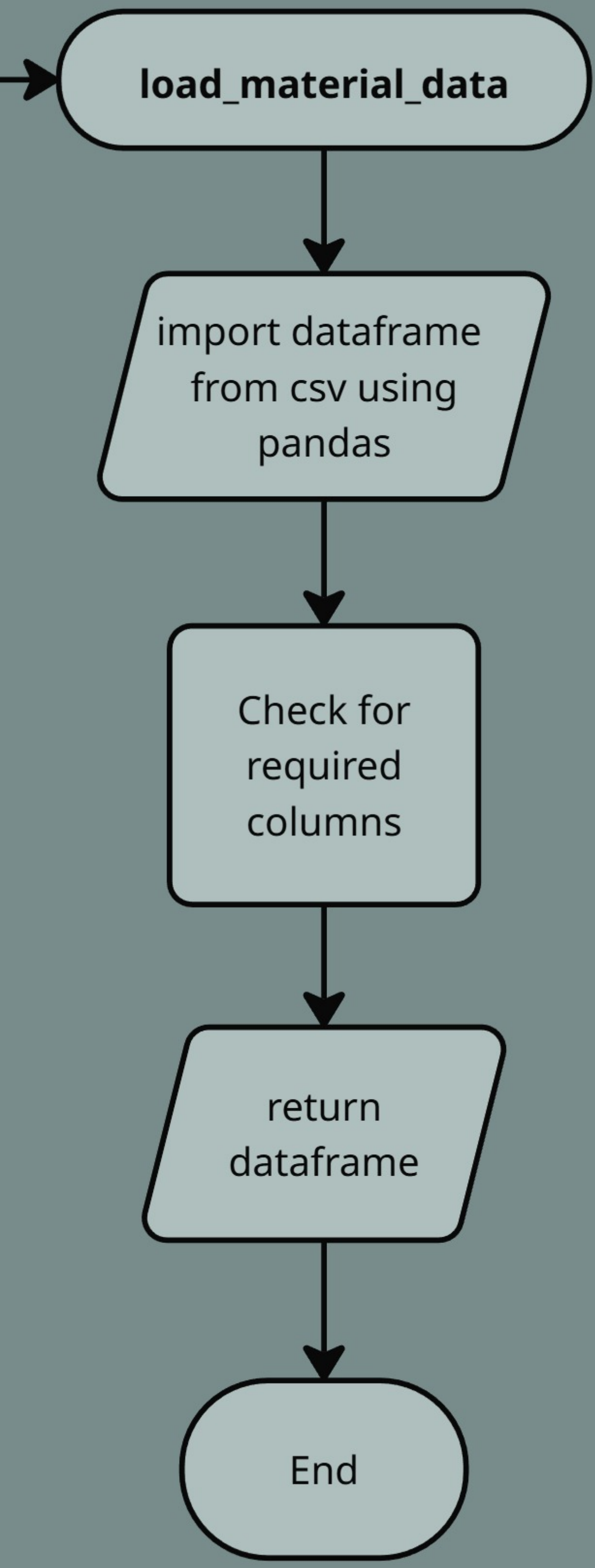
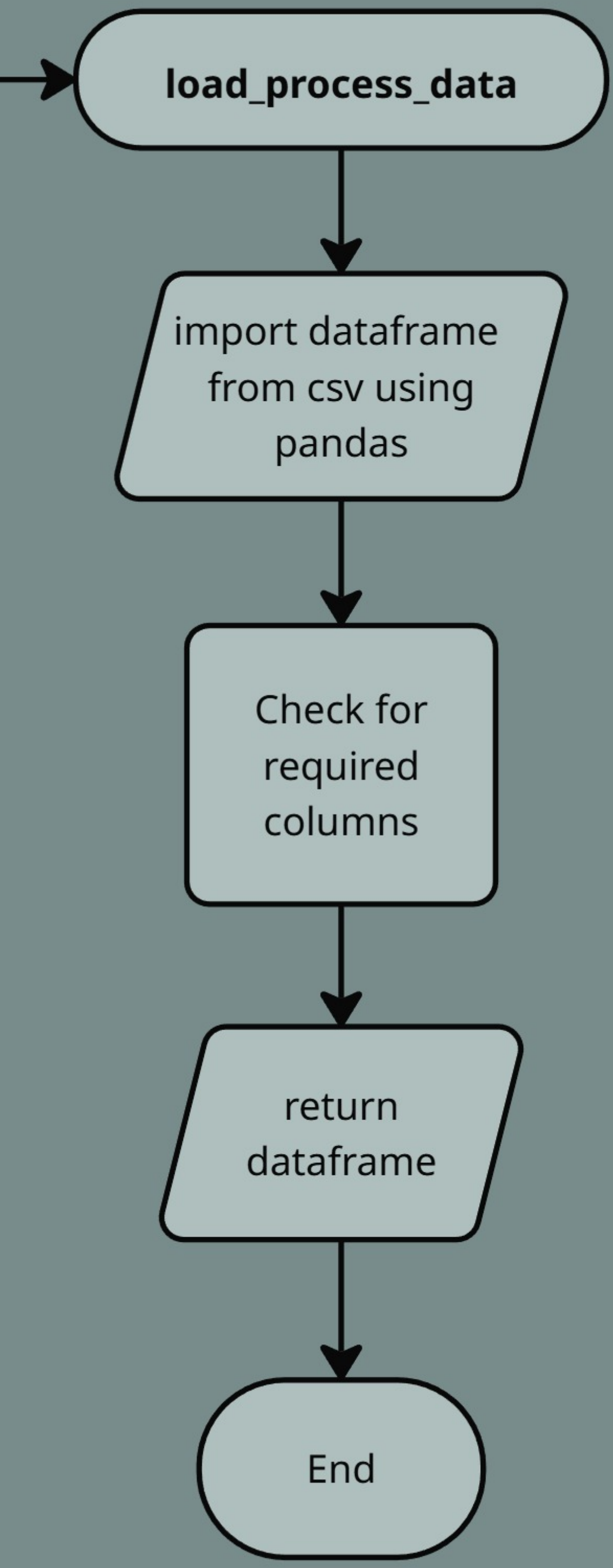
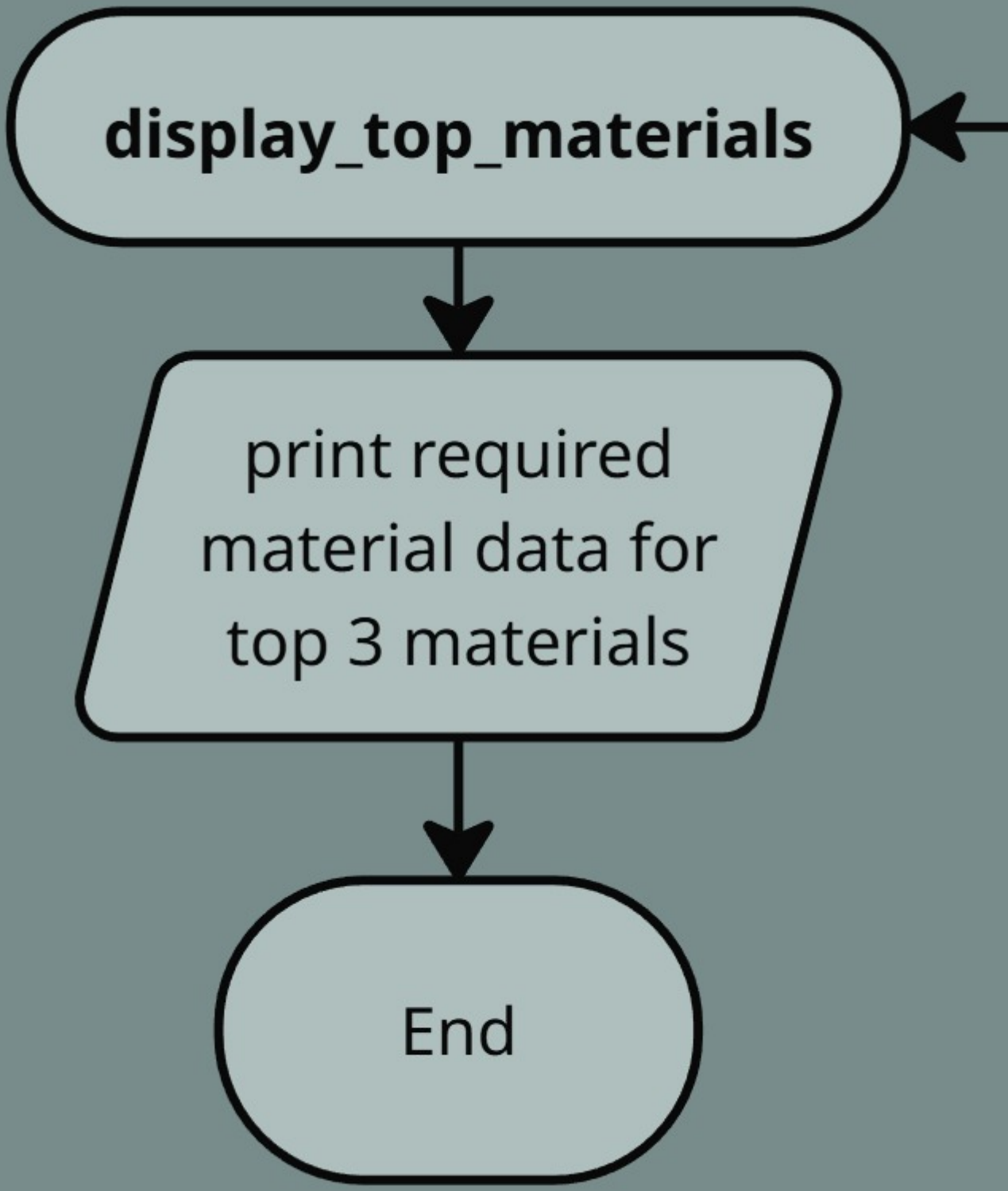


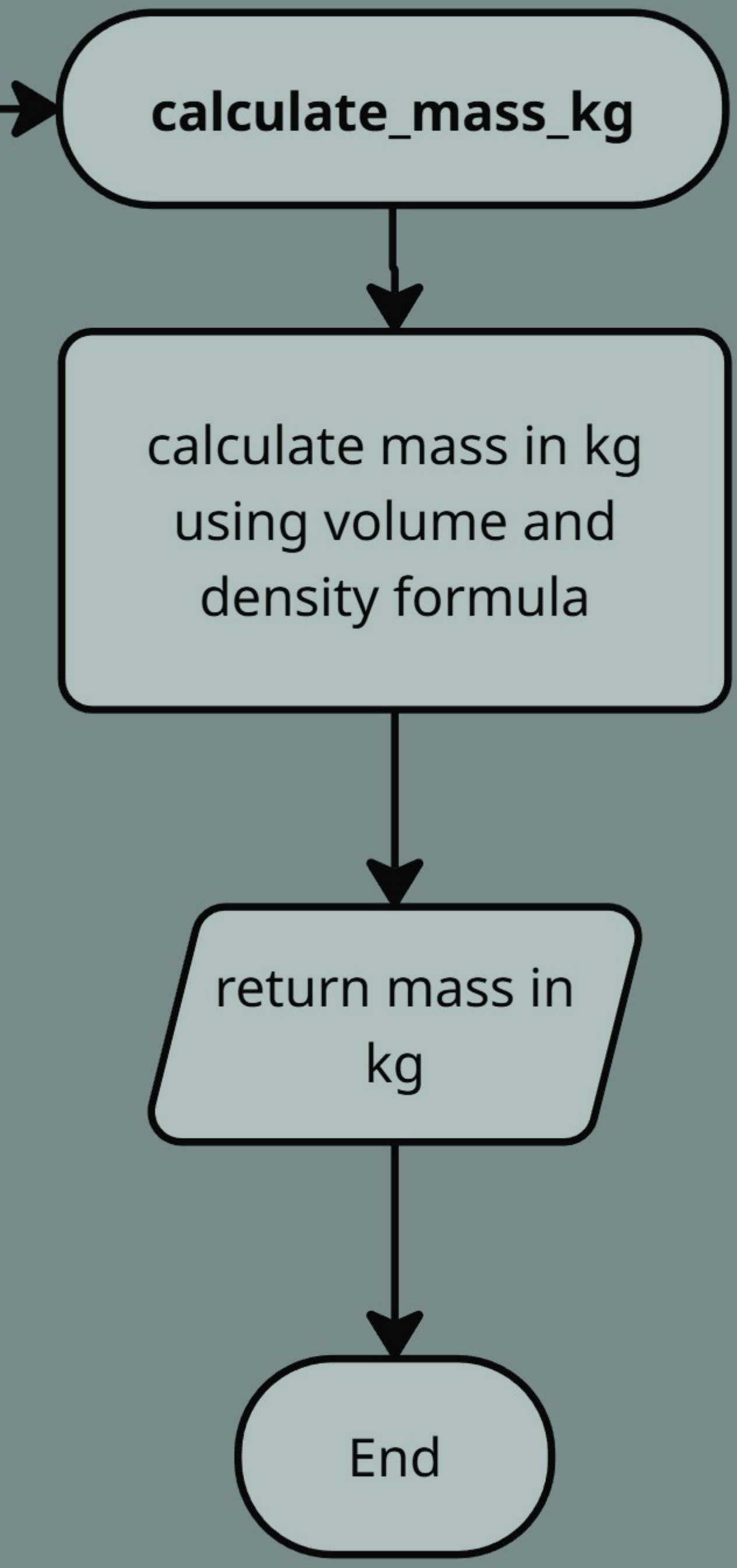

Start (main)
Create Outputs Directory if it doesn't exist
print opening message for program
prompt_material_constraints
output: constraints
load_material_data
input: relative path of csv file
output: dataframe of materials
train_material_model
input: dataframe of materials
output: material model
predict_materials
inputs: dataframe of materials, material model, constraints
output: sorted dataframe of materials in order of recommendation
no predictions possible?
True
Print Error Message
End
display_top_materials
input: sorted material dataframe
prompt_select_material
input: top 3 materials dataframe
output: chosen material data
print selected material data
prompt_part_specs
output: part specs dataframe
add material data to part specs dataframe
load_process_data
input: relative path of csv file
output: dataframe of processes
train_process_model
input: dataframe of processes
output: process model
predict_processes
inputs: dataframe of processes, process model, part specs
output: sorted dictionary of processes in order of recommendation, predicted process, confidence
create path to save process plot
plot_process_probabilities
inputs: sorted process dataframe, process plot path
outputs: saved plot file, displayed process plot
calculate_mass_kg
inputs: volume of part, density of material
output: mass of part in kg
estimate_total_cost
inputs: material cost per part, process cost per part, quantity of parts
output: total estimated cost
display_cost_breakdown
inputs: material name, process name, quantity, mass in kg, material cost per part, process cost per part, total cost
create path to save cost breakdown
save_cost_breakdown
inputs: material name, process name, quantity, mass in kg, material cost per part, process cost per part, total cost, breakdown path
print thank you message
End

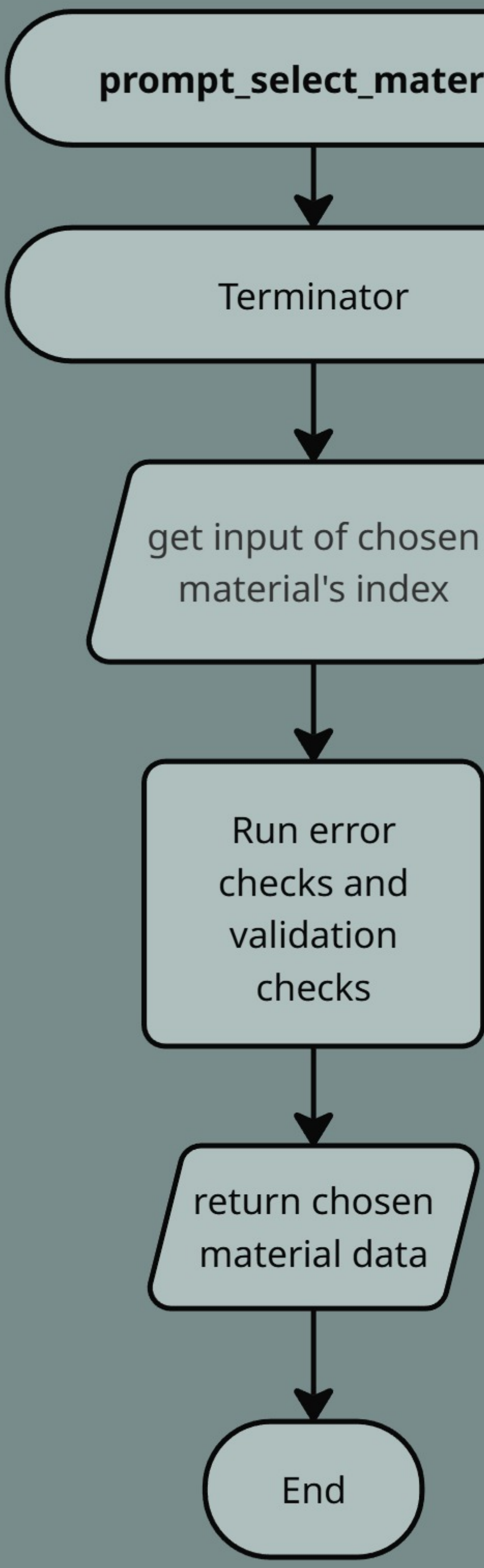

prompt_material_constraints
get input data of yield strength, density, and cost.
Run error checks and validation checks
return constraints
End

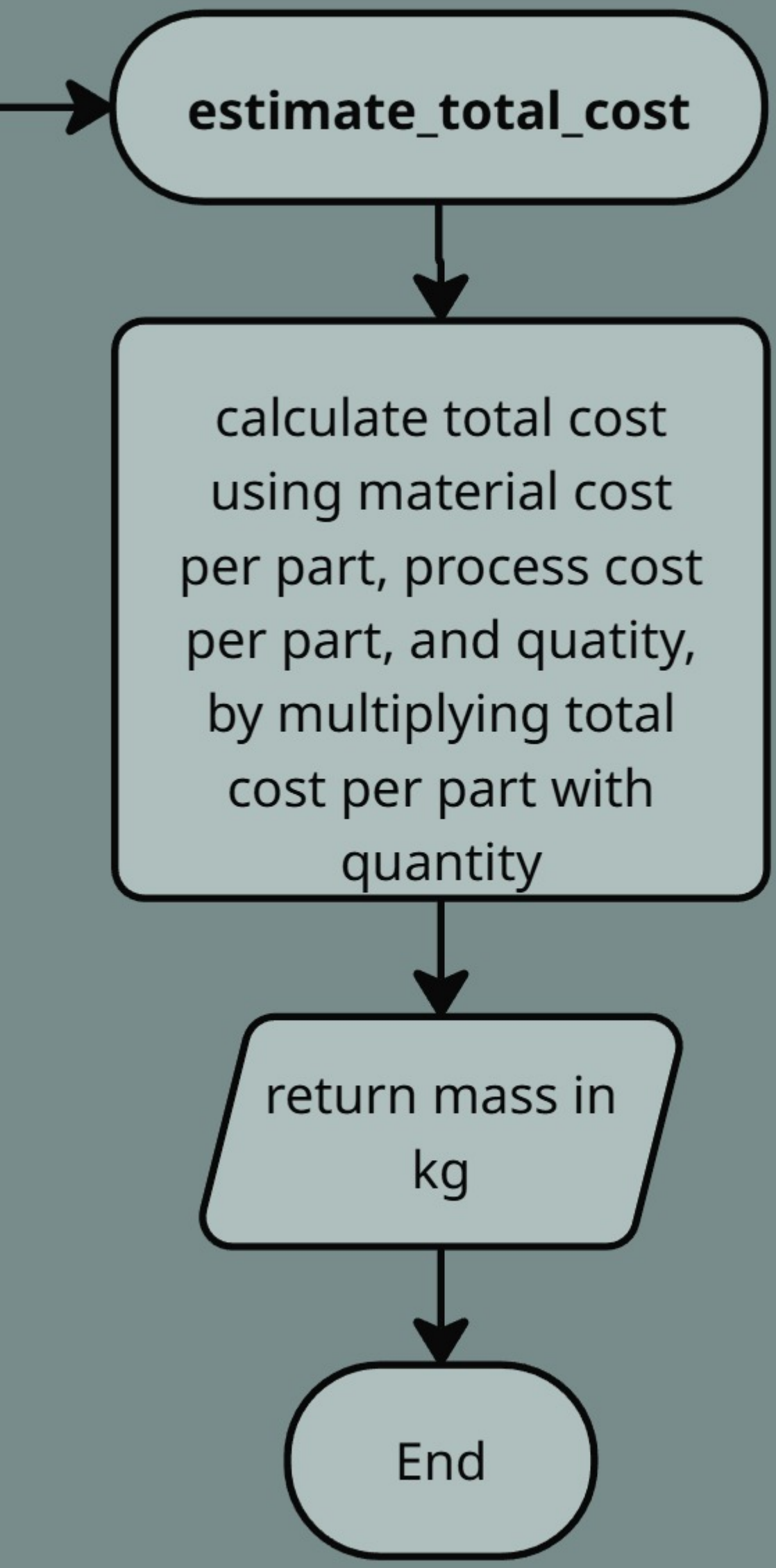

load_material_data
import dataframe from csv using pandas
Check for required columns
return dataframe
End

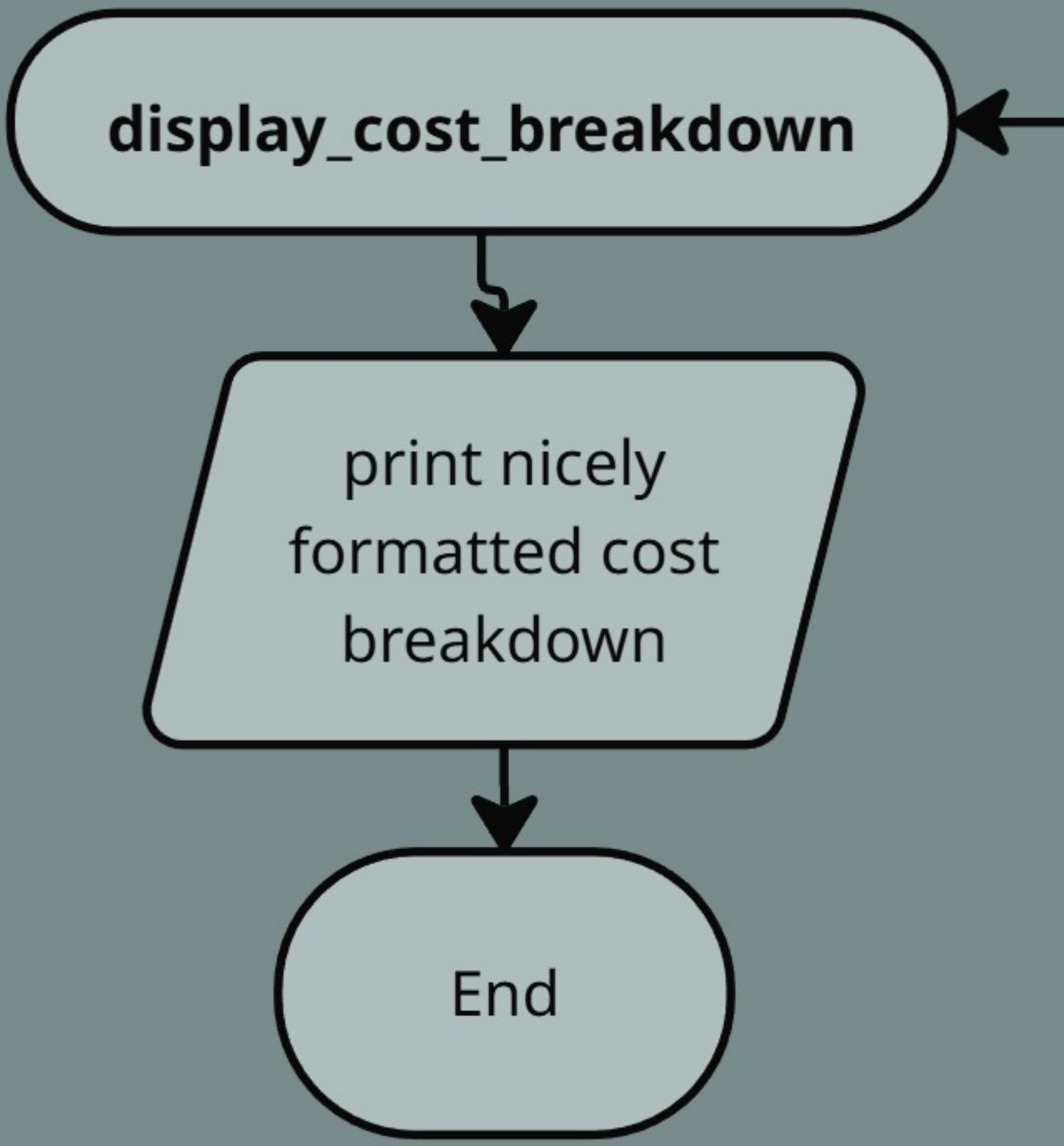

load_process_data
import dataframe from csv using pandas
Check for required columns
return dataframe
End

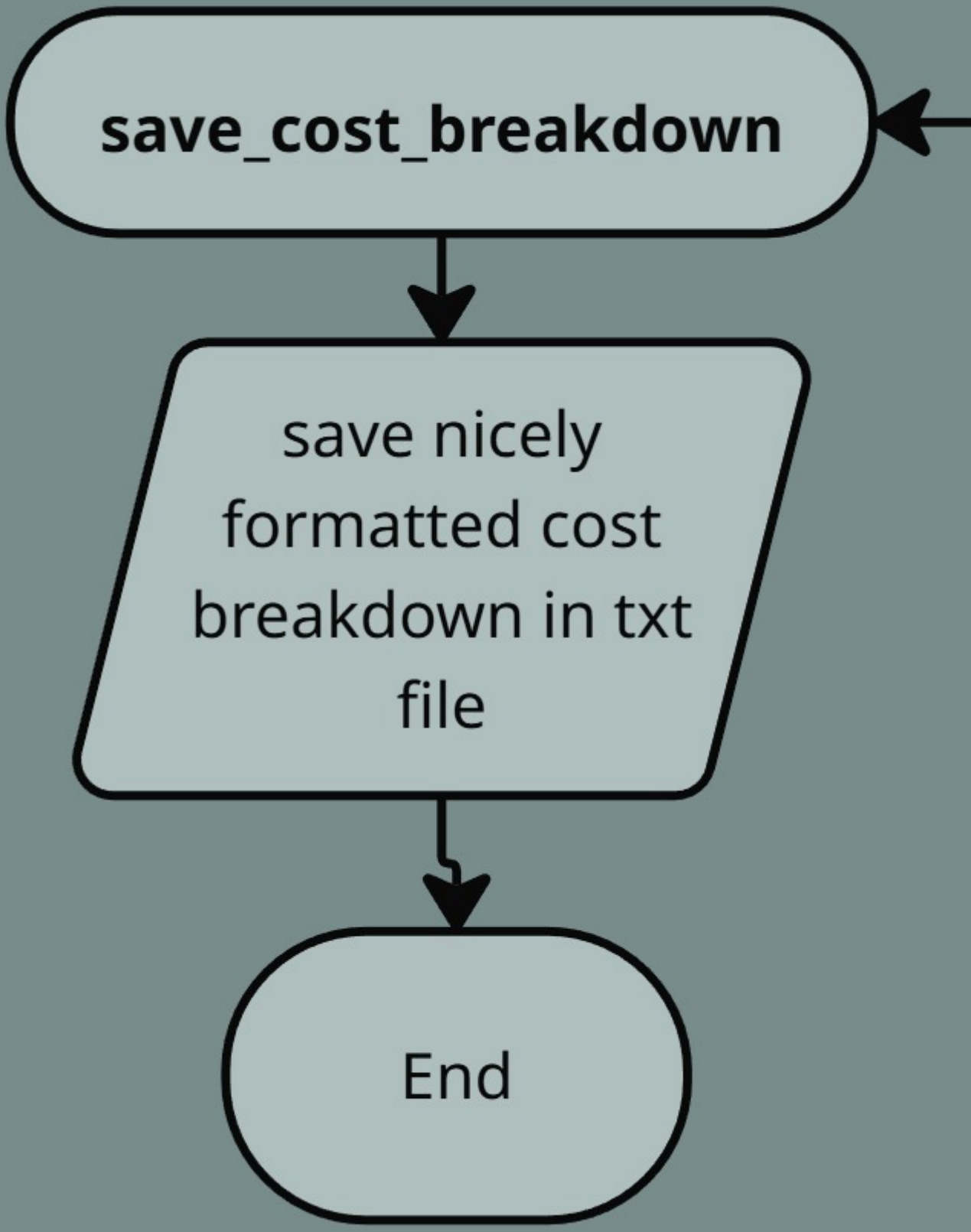

display_top_materials
print required material data for top 3 materials
End

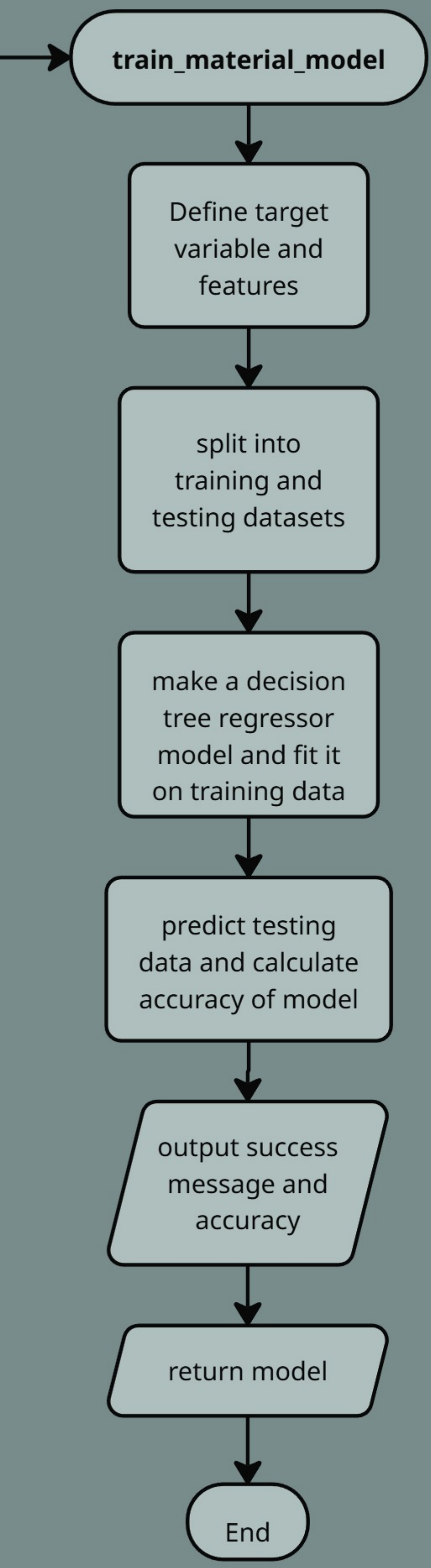

calculate_mass_kg
calculate mass in kg using volume and density formula
return mass in kg
End

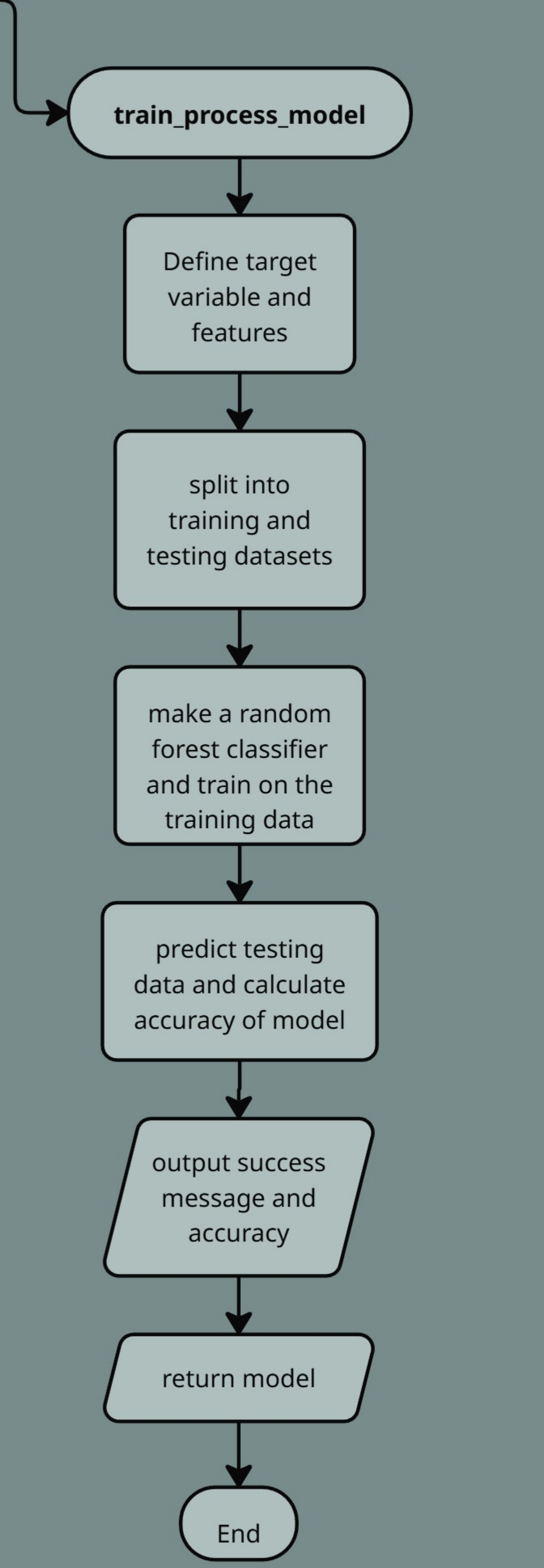

prompt_select_material
Terminator
get input of chosen material's index
Run error checks and validation checks
return chosen material data
End

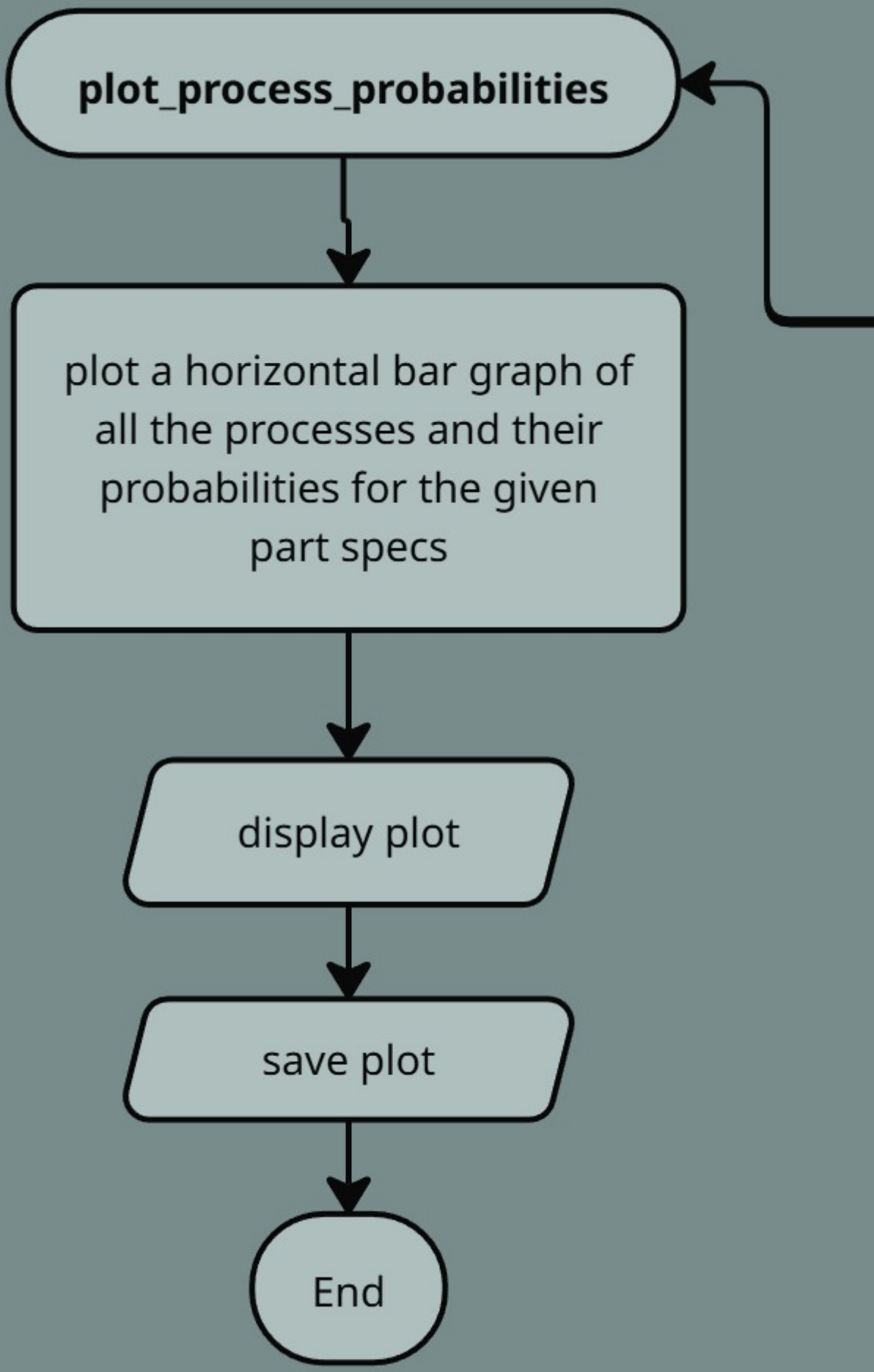

estimate_total_cost
calculate total cost using material cost per part, process cost per part, and quatity, by multiplying total cost per part with quantity
return mass in kg
End

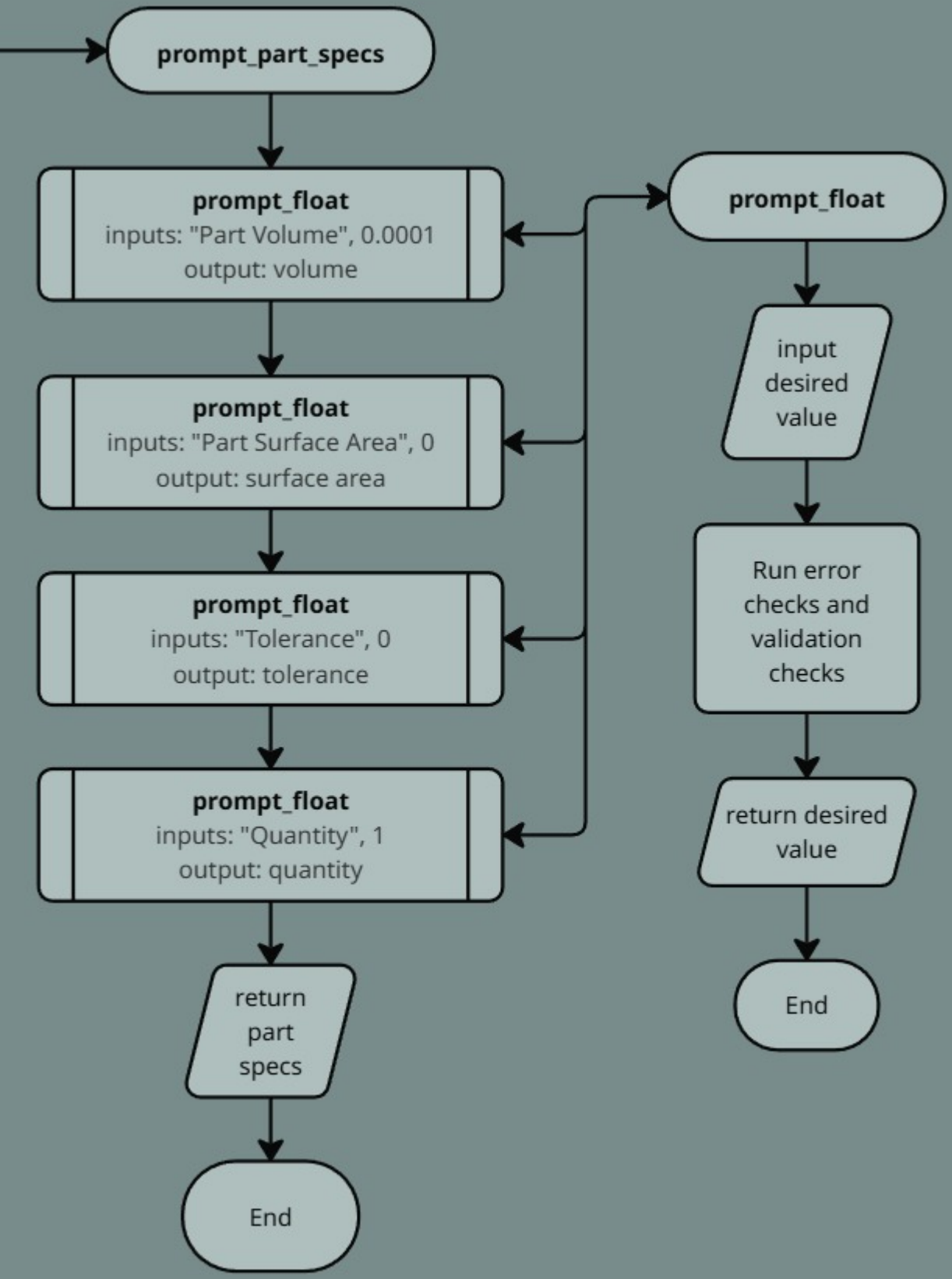

display_cost_breakdown
print nicely formatted cost breakdown
End

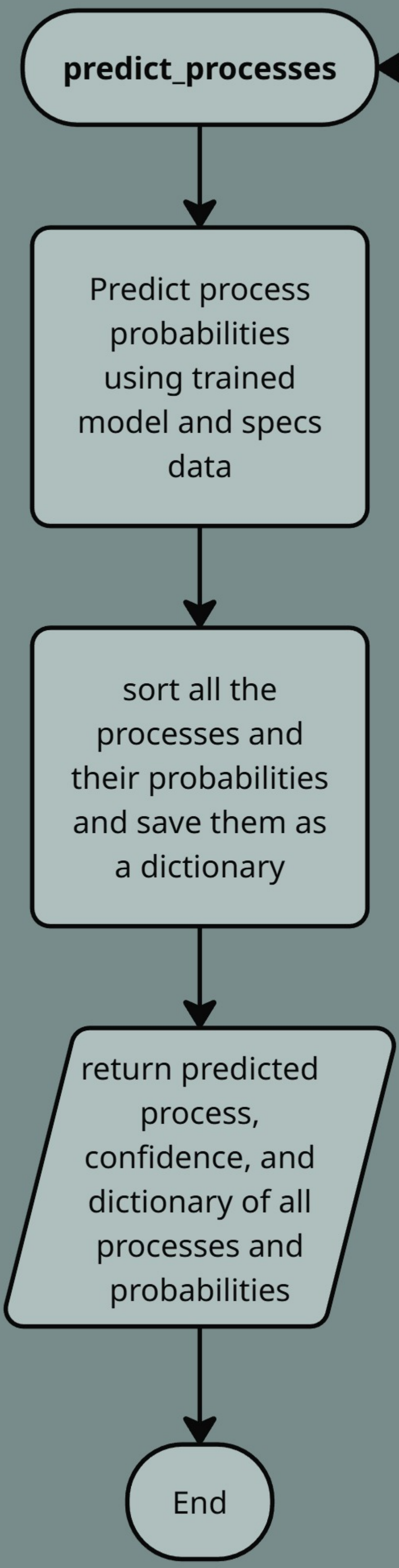

save_cost_breakdown
save nicely formatted cost breakdown in txt file
End

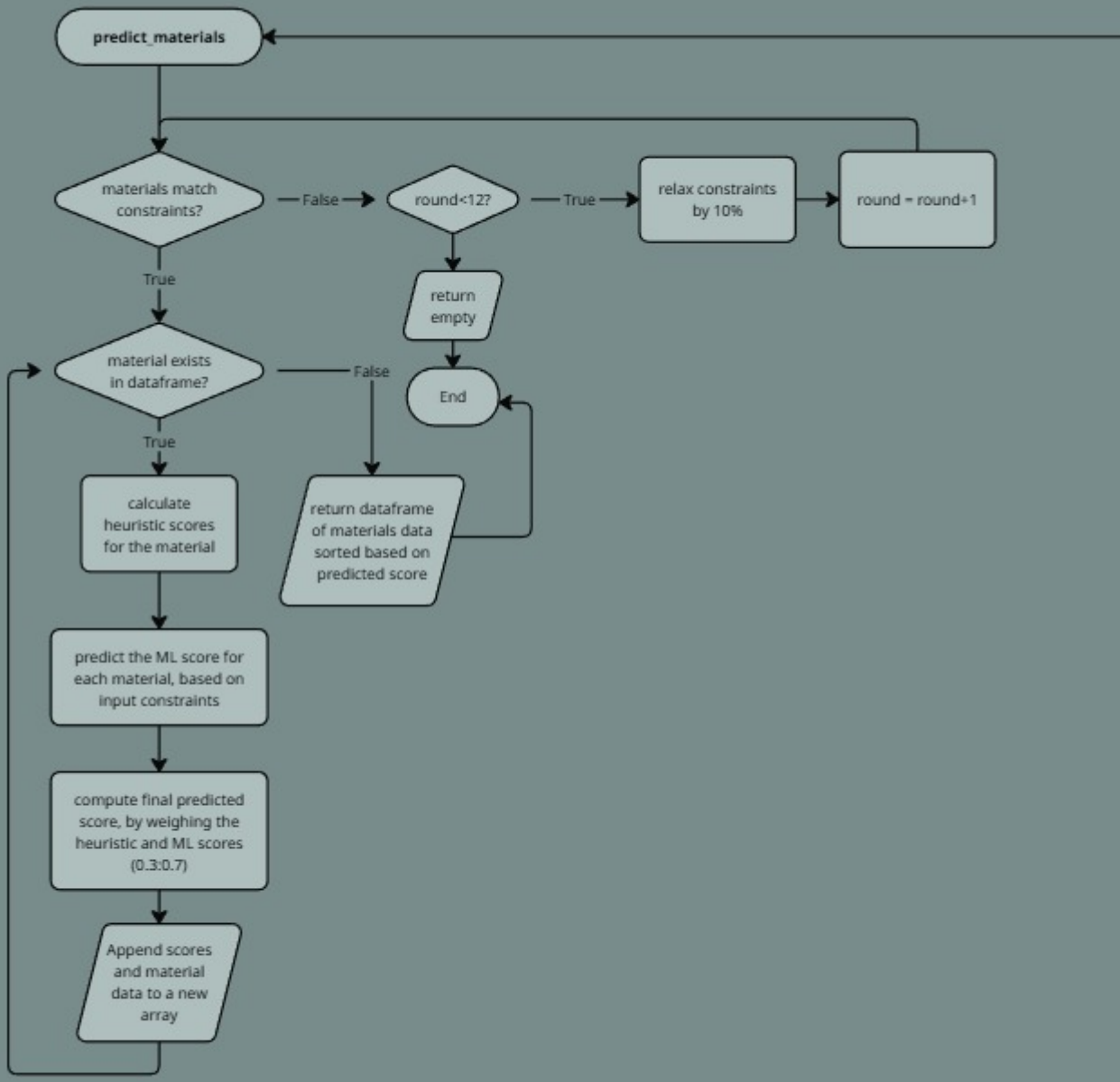

train_material_model
Define target variable and features
split into training and testing datasets
make a decision tree regressor model and fit it on training data
predict testing data and calculate accuracy of model
output success message and accuracy
return model
End


train_process_model
Define target variable and features
split into training and testing datasets
make a random forest classifier and train on the training data
predict testing data and calculate accuracy of model
output success message and accuracy
return model
End


plot_process_probabilities
plot a horizontal bar graph of all the processes and their probabilities for the given part specs
display plot
save plot
End


prompt_part_specs
prompt_float
inputs: "Part Volume", 0.0001
output: volume
prompt_float
inputs: "Part Surface Area", 0
output: surface area
prompt_float
inputs: "Tolerance", 0
output: tolerance
prompt_float
inputs: "Quantity", 1
output: quantity
return part specs
End
prompt_float
input desired value
Run error checks and validation checks
return desired value
End


predict_processes
Predict process probabilities using trained model and specs data
sort all the processes and their probabilities and save them as a dictionary
return predicted process, confidence, and dictionary of all processes and probabilities
End


predict_materials
materials match constraints?
False
round<12?
True
relax constraints by 10%
round = round+1
True
return empty
End
material exists in dataframe?
False
True
calculate heuristic scores for the material
return dataframe of materials data sorted based on predicted score
predict the ML score for each material, based on input constraints
compute final predicted score, by weighing the heuristic and ML scores (0.3:0.7)
Append scores and material data to a new array